

Analysis of viscoelastic PTT and Giesekus flows around sharp corner with slip boundary conditions

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1 Introduction

2 Mathematical formulation

The governing equations considered in this study comprise the continuity, momentum, and viscoelastic constitutive equations, given respectively by:

$$\begin{aligned}\nabla \cdot \mathbf{v} &= 0, \\ \frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla) \mathbf{v} &= -\nabla p + \frac{\beta}{\text{Re}} \nabla^2 \mathbf{v} + \frac{1}{\text{Re}} \nabla \cdot \boldsymbol{\tau}, \\ \boldsymbol{\tau} + \text{Wi} \overset{\nabla}{\boldsymbol{\tau}} + \frac{\epsilon}{1 - \beta} f(\boldsymbol{\tau}) &= 2(1 - \beta) \mathbf{D},\end{aligned}\tag{1}$$

where $\mathbf{v}$ is the fluid velocity vector, p is the pressure, $\boldsymbol{\tau}$ is the extra (viscoelastic) stress tensor, and $\mathbf{D} = \frac{1}{2} [\nabla \mathbf{v} + (\nabla \mathbf{v})^T]$ is the rate-of-deformation tensor. The parameter $\beta = \mu_s/\mu_0$ represents the viscosity ratio, defined as the ratio of the solvent viscosity μ_s to the total viscosity $\mu_0 = \mu_s + \mu_p$, where μ_p is the polymeric viscosity contribution.

The dimensionless groups are defined as follows:

- $\text{Re} = \frac{\rho UL}{\mu_0}$ is the Reynolds number, where ρ is the fluid density, while L and U denote the characteristic length and velocity scales, respectively.
- $\text{Wi} = \frac{\lambda U}{L}$ is the Weissenberg number, with λ representing the fluid relaxation time.

Furthermore, $f(\boldsymbol{\tau})$ is a model-dependent function that specifies the viscoelastic constitutive framework (e.g., Oldroyd-B, PTT, or Giesekus), and ϵ is the corresponding constitutive model coefficient.

To model fluid-wall interactions, we consider the *Navier-slip* boundary condition, expressed as:

$$\mathbf{v} \cdot \mathbf{t} = \kappa (\mathbf{t} \cdot \mathbf{n} : \boldsymbol{\tau}),$$

where κ is the dimensionless slip coefficient, $\mathbf{t}$ is the unit tangent vector, and $\mathbf{n}$ is the unit inward normal vector to the wall. Alternatively, this slip condition can be recast into the following equivalent form based on the rate-of-deformation tensor:

$$\mathbf{v}_{\text{wall}} = 2\kappa \left\{ \mathbf{n}^\top \cdot \mathbf{D} - [(\mathbf{n}^\top \cdot \mathbf{D}) \cdot \mathbf{n}] \mathbf{n} \right\}.\tag{2}$$

Table 1: Slope estimation for the stresses of PTT and Giesekus models.

κ_0	PTT			Giesekus		
	τ_{xx}	τ_{xy}	τ_{yy}	τ_{xx}	τ_{xy}	τ_{yy}
0	-0.348162	-0.371320	-0.444514	-0.3181	-0.3318	-0.4259
0.1	-0.369645	-0.390421	-0.460150	-0.3445	-0.3558	-0.4454
1	-0.372527	-0.391020	-0.438184	-0.5391	-0.5408	-0.6628
5	-0.576256	-0.571181	-1.866948	-	-	-
10	-0.528200	-0.526480	-0.575483	-0.5320	-0.5292	-0.5918
15	-0.480144	-0.478955	-0.502624	-	-	-
20	-0.438838	-0.437901	-0.453447	-	-	-
30	-0.373990	-0.373311	-0.382457	-	-	-
40	-0.325991	-0.325449	-0.331909	-	-	-
50	-0.289091	-0.288636	-0.293621	-	-	-
100	-0.186409	-0.186152	-0.188988	-0.1916	-0.1912	-0.1946
1000	-0.025363	-0.025309	-0.025543	-	-	-

3 Stress integration along streamlines

The estimated slopes for the stress components corresponding to the PTT and Giesekus models are summarized in Table 1.

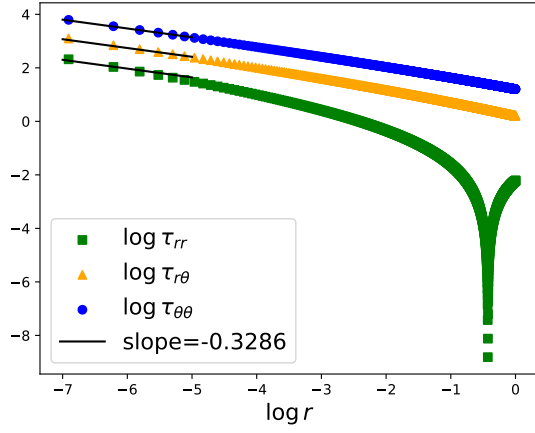
Figure 1 presents the stress profiles (τ_{rr} , $\tau_{r\theta}$, and $\tau_{\theta\theta}$) for the PTT model in polar coordinates along the streamlines intersecting the radial domain $r \in [0, 1]$ at $\theta = \frac{\pi}{2}$. The calculated slopes are consistent with the values reported in Table 1. Notably, for the largest slip coefficient ($\kappa_0 = 100$), the stresses begin to diminish. Additionally, Figure 1(f) displays the $\tau_{\theta\theta}$ stress component profiles across all analyzed slip coefficients to highlight their comparative behavior.

Figure 2 illustrates the stress profiles for the Giesekus model along the line $\theta = \frac{\pi}{2}$ for $r \in [0, 1]$. The results qualitatively mirror those obtained for the PTT model in Figure 1; specifically, both models exhibit a distinct reduction in the stress singularity at the largest slip coefficient, $\kappa_0 = 100$.

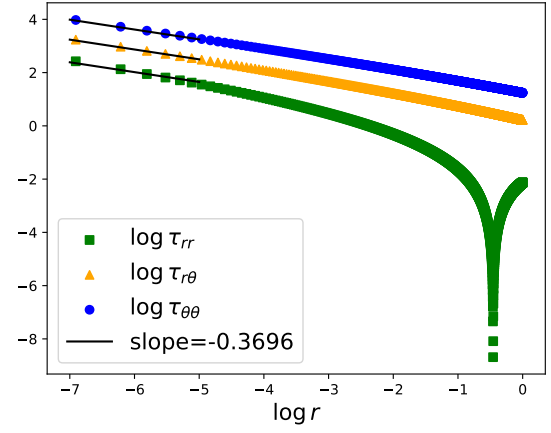
4 Numerical results

Figure 3 presents stress profiles analogous to those in Figures 1 and 2, but here obtained from simulations of the full set of governing equations [?]. The numerical results successfully capture the asymptotic slope for small values of κ_0 , corresponding to conditions close to the no-slip boundary. Conversely, as κ_0 increases, the τ_{xy} and τ_{yy} stress components do not properly align with the expected theoretical slope.

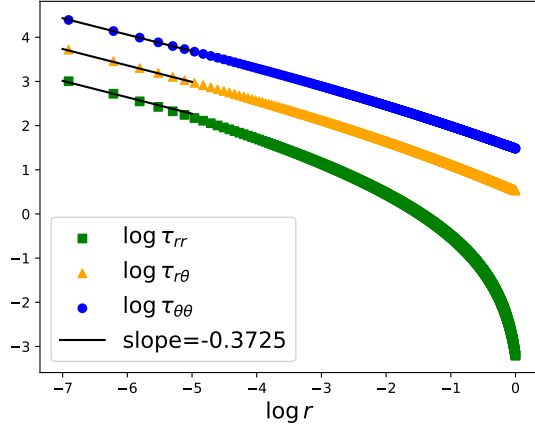
References



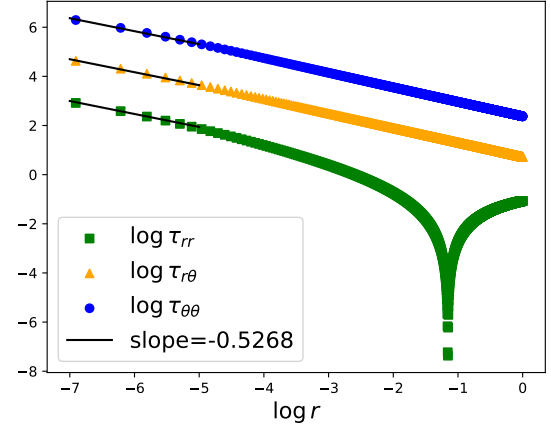
(a) $\kappa_0 = 0$



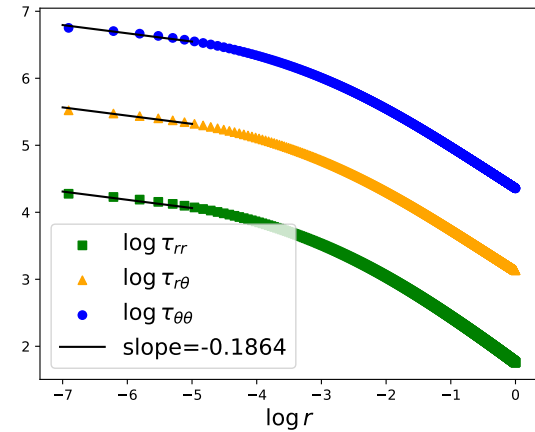
(b) $\kappa_0 = 0.1$



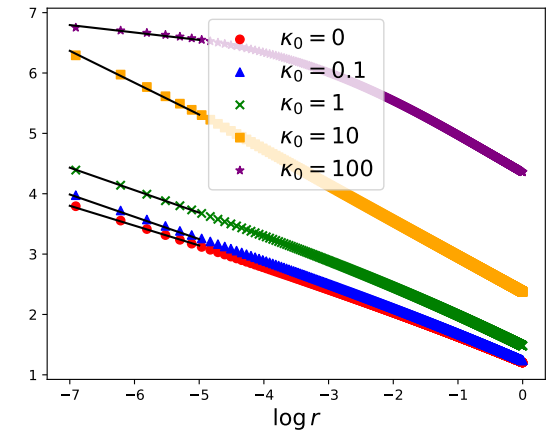
(c) $\kappa_0 = 1$



(d) $\kappa_0 = 10$

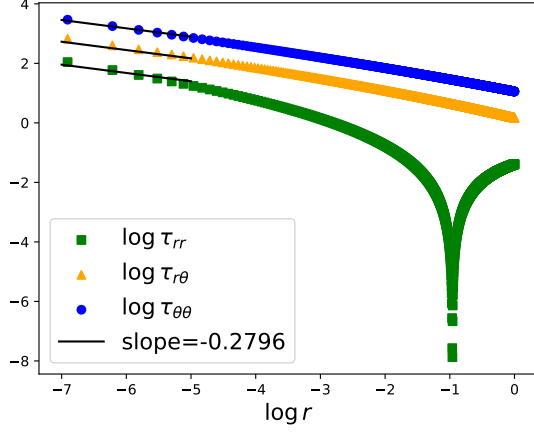


(e) $\kappa_0 = 100$

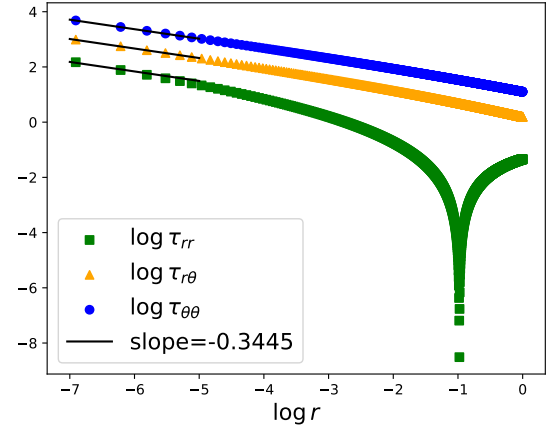


(f) $\tau_{\theta\theta}$ for all κ_0

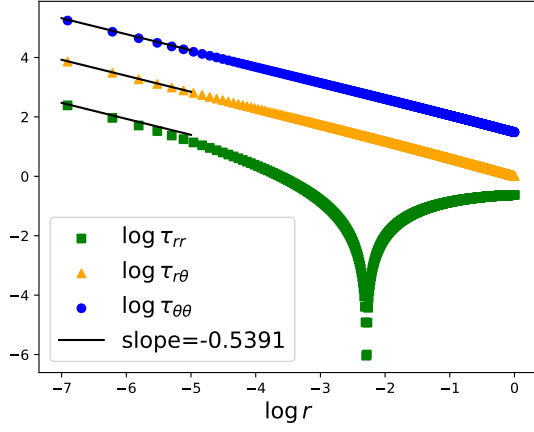
Figure 1: Stresses profile along the radius $\theta = \frac{\pi}{2}$ using the integration along streamlines for the PTT viscoelastic model.



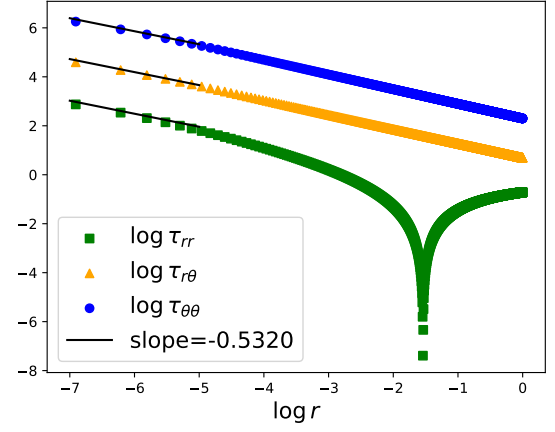
(a) $\kappa_0 = 0$



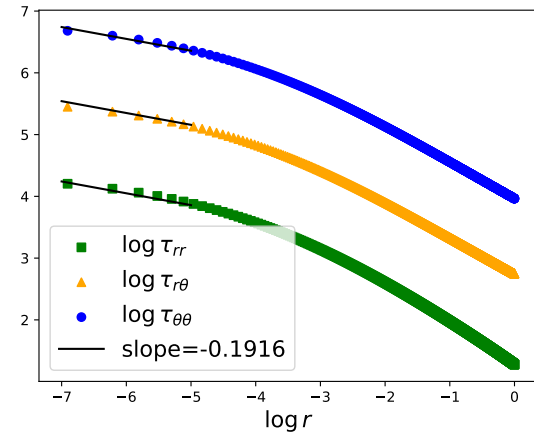
(b) $\kappa_0 = 0.1$



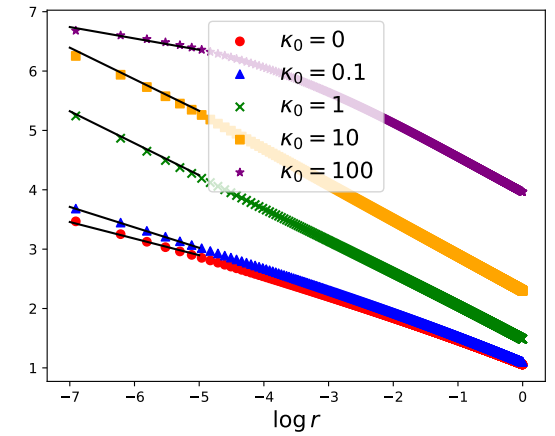
(c) $\kappa_0 = 1$



(d) $\kappa_0 = 10$

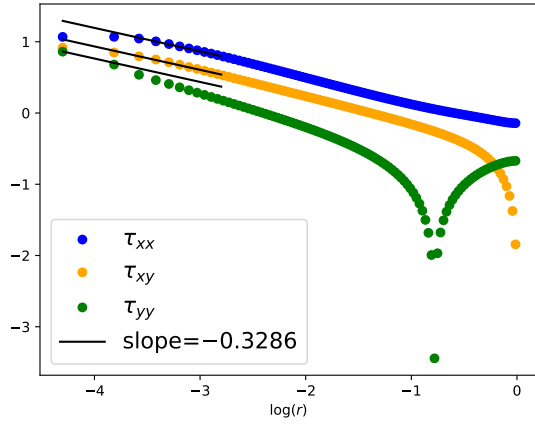


(e) $\kappa_0 = 100$

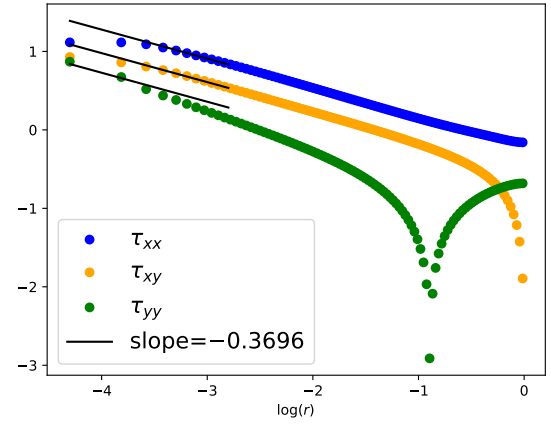


(f) $\tau_{\theta\theta}$ for all κ_0

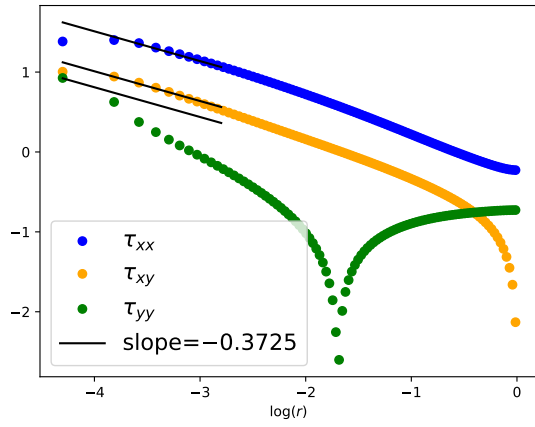
Figure 2: Stresses profile along the radius $\theta = \frac{\pi}{2}$ using the integration along streamlines for the Giesekus viscoelastic model.



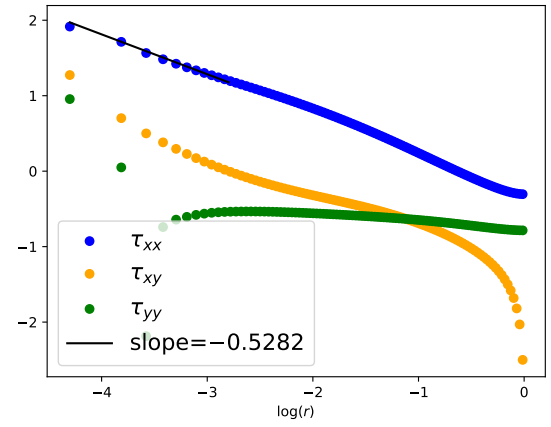
(a) $\kappa_0 = 0$



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Figure 3: Stresses profile along the radius $\theta = \frac{\pi}{2}$.